Tennis Ball Tracker PDR Documentation

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Amelia Fries, Cooper Felle, Jackson Charnock, Matthew Chang, Mike Herr, William Battaglia

Rochester Institute of Technology
CPET 563: Embedded Systems II Lab

Contributions

- Cooper Felle
  - MATLAB GUI
  - Ball detection pipeline
  - Blender environment/simulation
- Jackson Charnock
  - Camera Research
  - GUI Development
  - LED Visualization
  - Volley Tracking
  - Documentation
- Amelia Fries
  - Snickerdoodle integration/development
  - Coefficient of restitution
  - IN/OUT Net Logic
- Mike Herr
  - Blender Interfacing/Setup
  - C Development
- Matthew Chang
  - Web server development
  - HDL Interfacing
  - Python/Linux
- William Battaglia
  - Serve Tracking
  - Instant Replay Feature
  - Documentation

The goal of this project is to develop a system capable of accurately determining the 3-D position of a tennis ball by using stereo vision techniques. The system will support real-time tracking and decision-making, such as determining whether a ball lands in or out of bounds, while also maintaining high accuracy under varying environmental conditions (i.e. weather). Existing tennis tracking systems rely on complex multi-camera configurations and are susceptible to environmental disturbances such as wind-induced camera movement, which introduces significant positional error. In response to this, the project proposes a stereo vision-based approach that reduces system complexity while still maintaining high accuracy through computational methods and controlled simulation.

The system is built upon an integrated hardware and software architecture designed to support both development and real-time operation. The hardware platform uses a Snickerdoodle SoC, which combines ARM processing cores with FPGA logic to enable accelerated image processing and real-time decision making [7], [8]. Camera inputs are captured as stereo image pairs and processed through a MATLAB-based graphical user interface (GUI), while Blender is used to generate a virtual tennis court environment for controlled testing and validation. A combination of python/HDL/C is used as an intermediary layer to facilitate communication between the simulation environment, processing algorithms, and hardware components. This hybrid architecture allows for easier debugging and performance evaluation before transitioning to a fully deployed system.

The core image-processing pipeline begins with gathering the stereo image, where synchronized left and right camera frames are captured. These images are then processed to detect the tennis ball and determine its pixel coordinates within each frame. The difference in horizontal pixel position between the left and right images, disparity, is used to estimate depth. Using known camera parameters such as focal length and baseline distance between cameras, the system computes the 3-D position of the ball through triangulation [5], [6]. These coordinates are then used to determine whether the ball is within court boundaries, and the system outputs both a visual and hardware-based indicator of the result.

One critical piece of the system is the ball detection algorithm, which must balance accuracy and efficiency. Several detection methods were evaluated, including: grayscale

thresholding, YCbCr color-based detection, and circle detection using the Hough Transform. Grayscale thresholding works by converting an image to intensity values and segmenting pixels based on a selected threshold [1], [2]. This method is computationally efficient and suitable for real-time applications, but is also highly sensitive to lighting conditions and may incorrectly detect court lines/other bright objects like the ball. YCbCr color detection improves upon this by separating luminance from chrominance, allowing the system to isolate the characteristic yellow-green color of a tennis ball [3]. This reduces false detections caused by white court markings and varying lighting conditions, but still produces generalized regions rather than precise circle boundaries.

To achieve higher accuracy, circle detection is used to identify the geometric shape of the tennis ball. This method analyzes edge information and uses a voting process to determine the most likely circle parameters, providing both center and radius of the detected objects. Although this approach is significantly more computationally intensive, it does offer a more accurate representation of the ball's position [4]. Based on the strengths and weaknesses of each method, a hybrid detection pipeline was implemented: the system first applies a coarse detection method, such as grayscale thresholding or YCBCr, to identify a region of interest. This region is then refined using the circle detection to accurately determine the ball's centroid. This approach should, in theory, reduce computational load while maintaining accuracy, which is critical for the system.

The accuracy of the system is heavily dependent on precise centroid detection due to the nature of stereo triangulation. Depth is calculated using the relationship between disparity, focal length, and camera baseline, where depth is inversely proportional to the disparity. Essentially, as the disparity between the two camera images decreases, particularly for objects farther away, small errors in pixel localization can result in large errors in depth estimation [5], [6]. This emphasizes the importance of accurate image processing and highlights the need for the robust detection algorithms mentioned. Additionally, the system calculates the full 3-D position by projecting pixel coordinates into world space using camera calibration parameters, including the principal point of each image.

Camera configuration plays a significant role in system performance and design tradeoffs. A two-camera stereo system offers simplicity, lower cost, and easier calibration, making it easier for initial implementation and controlled environments; however, this configuration is more susceptible to occlusion and may experience reduced accuracy at longer distances due to decreased disparity. To address these limitations, a four-camera configuration using two stereo pairs is also considered. This approach provides multiple viewpoints, increased redundancy, and improved robustness. The tradeoff for this increased accuracy is higher system complexity, greater computational requirements, and increased cost [10] - [12].

System validation is performed through a series of controlled tests within the simulation environment. One test involves keeping the camera positions fixed while moving the ball through a range of distances, typically from three to five meters (while the ball will likely be further on the court, this provides a depth-change estimate), and comparing the estimated position to the known ground truth. Another test involves keeping the ball stationary while introducing simulated camera movement to evaluate the impact of weather on accuracy.

To further evaluate the performance of each detection method, both timing and positional accuracy were measured for an image of a white ball at a fixed location: (0, 0, 0). The results are summarized in the table below:

Timing Analysis (ms):

GrayThresh (Fig. 1-2)	YCbCrNeutral (Fig. 3-4)	Found-Circle (Fig. 5-6)
15.6 - 16.9	41.0 - 44.8	200.0 - 278.1

X,Y,Z Position:

GrayThresh (Fig. 1-2)	YCbCrNeutral (Fig. 3-4)	Found-Circle (Fig. 5-6)
(-0.036, 0.0311, 0.410)	(-0.0357, 0.0334, 0.4258)	(-0.0330, 0.0369, 0.4094)

The grayscale thresholding method demonstrated the fastest execution time compared to YCbCr and the circles method, respectively. In terms of positional accuracy, all three methods produced relatively similar 3-D estimates, with minor variations in the computed coordinates. These results

indicate that, although the circle detection method may provide a more geometrically accurate centroid, the overall improvement in position estimation is marginal compared to the increase in its computational cost. This reinforces the decision to implement a hybrid pipeline, where faster methods are used for coarse detection and circle detection is used for refinement.

In addition to position tracking, the system has several functional features relevant to tennis gameplay analysis. One example of this is the calculation of the coefficient of restitution, which measures the bounciness of a court's surface by comparing the ball's velocity before versus after impact. The system also includes logic for determining whether a serve or volley is in or out of bounds by comparing the ball's landing position to predefined court dimensions [9]. These decisions are communicated through both a graphical user interface and hardware outputs, such as LEDs controlled by the FPGA, where different colors indicate in/out conditions. The system also supports trajectory storage and replay functionality, allowing users to see the path of the ball, bounce locations, and final decisions within the simulation environment. The integration of hardware acceleration through the FPGA ensures that the system can perform these features and meet real-time performance requirements, particularly when processing high-frame-rate camera inputs [7].

From a cost perspective, the system expenses include: cameras, processing hardware, software licensing, and development time. High-quality cameras capable of at least 60 Hz operation are needed to capture fast-moving tennis balls accurately and the FPGA-based processing hardware adds to the cost but is justified by the need for real-time performance. Overall system costs are estimated to be in the range of approximately $30,000, including labor and equipment [10] - [13].

The project follows a well-structured development timeline, starting with initial implementation of the GUI and ball detection pipeline, followed by integration with the simulation environment and hardware components. Final stages include implementation of advanced features, system optimization, and debugging.

In summary, this project presents a high-level approach to tennis ball tracking using stereo vision, combining simulation, image processing, and hardware acceleration to achieve accurate and real-time performance. By leveraging a hybrid detection pipeline and considering

system tradeoffs, the design addresses many of the limitations found in existing solutions while maintaining scalability for future enhancements.

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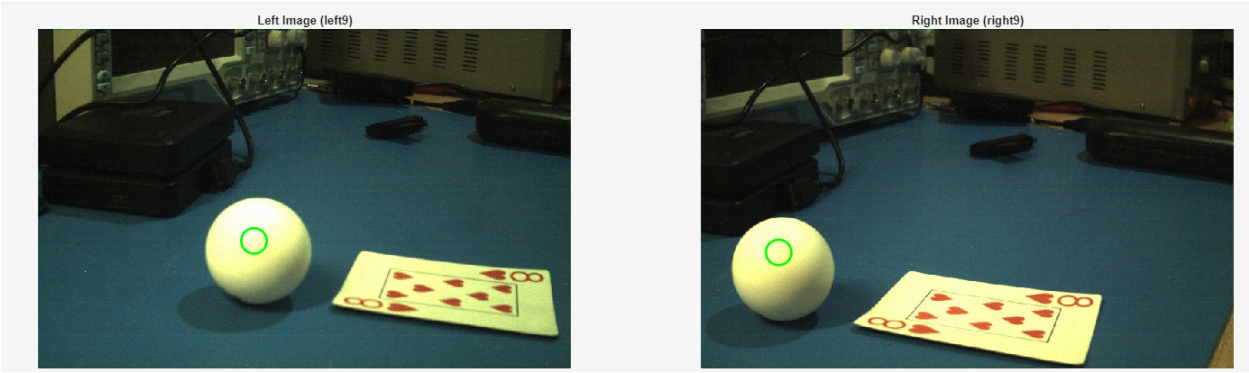

Fig. 1. GrayThresh detection showing thresholded segmentation and estimated centroid.

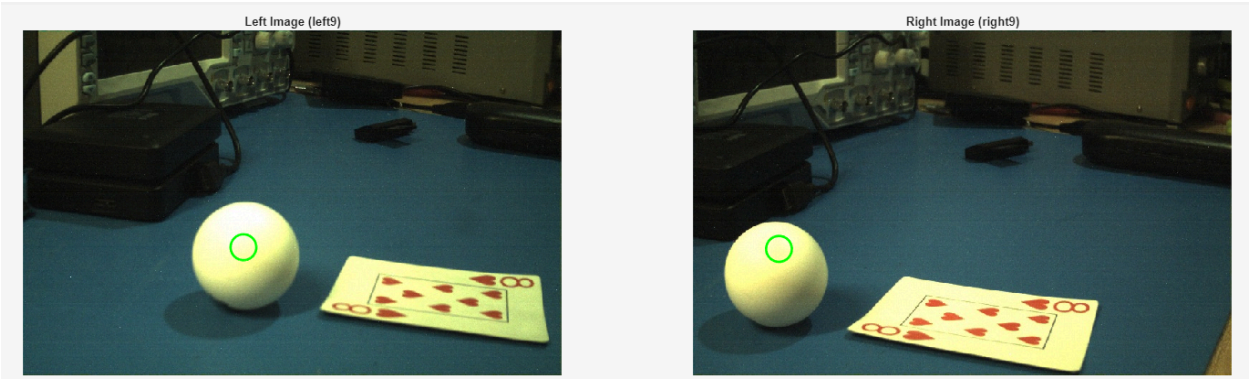

Fig. 2. YCbCr-based detection isolating the tennis ball color using chroma thresholding.

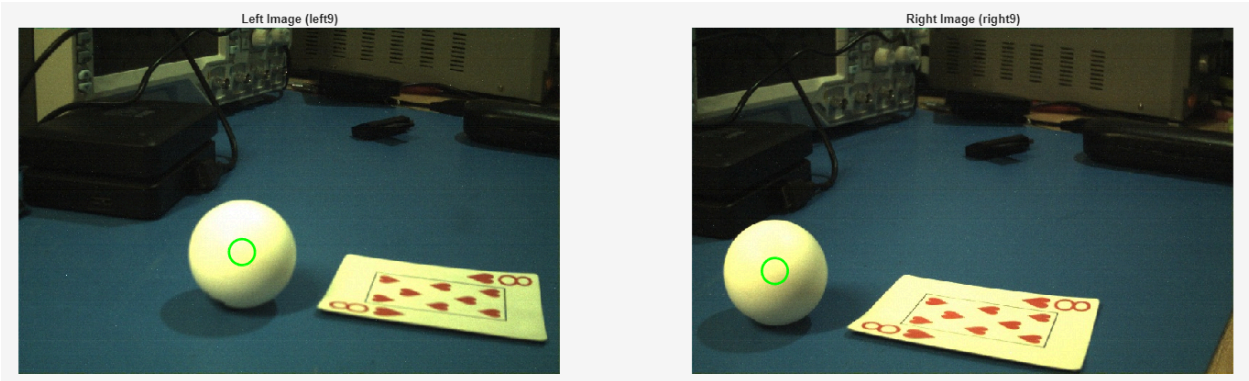

Fig. 3. Circle detection using Hough Transform identifying ball center and radius.